

Speaker Notes: From Job Adverts to National Accounts

Target Duration: 20 - 22 Minutes (Including Stakeholder's slides) **Audience:** NACD Division

Slide 1: Title Slide

Speaker : YOU

"Good afternoon, everyone. Thank you for having me.

Today, I want to discuss a fundamental challenge we face in modern economic measurement: how do we put a price tag on an intangible asset we cannot physically see?

Over the next 20 minutes, I'm going to walk you through how I used Artificial Intelligence to transform a massive dataset of unstructured job advertisements into a robust, National Accounts-ready framework for valuing 'Data' in the UK economy.

But before we get into the methodology of *how* I did this, I'm going to hand over to

Stakeholder's Name

to explain exactly *why* this is such a critical priority for the division right now."

Slide 2: Why are we doing this? (NEW SLIDE)

Speaker : STAKEHOLDER

(Note for you: Just advance the slide and let them speak. They will cover:)

- The upcoming SNA 2025 changes introducing "Data and Databases" as a new IPP asset.
- The specific definition of Data (digital format, economic benefit, recorded observations).
- The crucial point that data can be purchased OR **produced by businesses for their own use (own-account)**.

Slide 3: The Challenge & The Solution

Speaker : YOU

(Transition back to you)

"Thank you,

Stakeholder's Name

So, as we just heard, we are now mandated to measure 'own-account' data—data produced by businesses for their own use.

This presents a massive macroeconomic blind spot. You cannot look at a corporate balance sheet and see the exact monetary value of a bespoke, internal database. If we don't measure these data assets, we fundamentally underestimate the UK's Gross Value Added.

To solve this, I relied on a well-established accounting methodology: the 'Sum-of-Costs' approach. The OECD specifically recommends applying this existing method to internal data assets. The logic is simple: if we cannot observe a market price for a company's internal data, we value the asset based on what it cost the company to build it.

The primary cost of building data assets is human labor. If I could identify the workers building these assets and measure their compensation, I would have the foundational building blocks for capital valuation."

Slide 4: The Flaw in "Job Titles"

Speaker : YOU

"But finding these workers isn't as simple as pulling a list of standard occupational codes.

Let's look at our running example for today: **SOC 4114 - Data Entry Administrators**.

The reason I focus on Data Entry operators is that they are fundamentally seen as the people *producing* the data. This aligns perfectly with the 'own-account' data creation we just discussed. We have an exact SOC code for this, so in theory, we could just count every job advert under 4114 and call it a day.

But we cannot simply count all the jobs that belong to that SOC code.

Why? Because a job title is just a label. Many roles under that code might just involve basic typing or generic admin work that doesn't actually contribute to a valuable corporate data asset. Conversely, highly technical data production is happening under completely different job titles like 'Marketing Manager' or 'Logistics Coordinator'.

If we rely purely on static job titles or broad SOC codes, our estimates will be wildly inaccurate. We must measure actual technical capabilities. And to do that, I needed alternative data."

Slide 5: The Alternative Data Engine

Speaker : YOU

"Instead of relying on lagging corporate surveys, I went straight to the labor market.

I ingested 62.3 million raw UK job advertisements spanning the last five years. These historical adverts act as real-time employer 'wish lists', telling us exactly what technical skills are demanded on the ground.

To process this massive text corpus, I engineered a PySpark Natural Language Processing—or NLP—pipeline.

A critical feature of this pipeline is **Noun Chunk Extraction**. Instead of just counting single, isolated words which lose their meaning, the pipeline extracts complete grammatical 'chunks'. For example, it extracts the full phrase 'relational database management' rather than just the word 'database'. This allows the model to capture the exact technical context of what the employer is asking for."

Slide 6: The AI Filtration Pipeline

Speaker : YOU

"Of course, mining job adverts is incredibly noisy. To isolate genuine data skills, I built a strict, three-stage mathematical filter.

First is the 0.35 Cosine Semantic Boundary. The AI maps the extracted noun chunks into a 300-dimensional semantic space. I require a 0.35 minimum similarity score against the core concept of 'data'. This threshold is carefully calibrated: it permanently deletes unrelated jargon—like 'office catering'—while preserving functional tools like 'Python', which a stricter threshold would wrongfully reject.

Second is the 10.0x Relative Share rule. Semantic similarity isn't enough. The word 'Spreadsheet' scores highly, but basic Excel use doesn't equal capital formation. To survive, a skill must appear at least 10 times more frequently in specialized Anchor Occupations—like our Data Entry professionals—than in the baseline UK economy. This brutally filters out generic office noise.

Finally, the 0.45 Gold Standard Polish. Statistics can be tricked. If a firm mandates its data analysts wear a 'purple necktie', that phrase might mathematically spike. To prevent anomalies, surviving noun chunks are tested against undeniable industry pillars. If a phrase doesn't hit a strict 0.45 similarity to these pillars, it is permanently deleted."

Slide 7: Classification & Mapping

Speaker : YOU

"With a mathematically verified dictionary in hand, the pipeline scans the entire dataset to classify the workforce.

I set a highly conservative classification threshold: an advert must contain **three or more distinct dictionary terms** to be officially classified as 'Data-Intensive'.

But for National Accounts, workforce numbers are not enough. We have a structural problem: Job adverts use SOC—Occupational codes. But Supply and Use Tables use SIC—Industry codes.

To bridge this gap, I engineered a Probabilistic Crosswalk Matrix. Using demographic weights from the UK Census, I can take our classified SOC 4114 Data Entry workers and mathematically distribute them into their actual operating industries—40% to Information, 25% to Finance, and so on. Unstructured labor text is now officially mapped into structured industry sectors."

Slide 8: The Valuation Framework

Speaker : YOU

"This crosswalk allows the economics to finally take over.

Once our data workers are correctly allocated to their SIC industries, I link this workforce back to the ONS Supply and Use Tables. This isolates their specific Compensation of Employees (CoE)—the base wage cost of building the data.

But a worker's wage isn't the total value of the asset. We must account for physical tools, operational overheads, and operating surplus.

To do this, I apply our Alpha markup multiplier, derived directly from ONS Input-Output tables. By multiplying the Compensation of Employees by the Alpha markup, raw wage data is converted into a scaled, macroeconomic valuation of total capital investment in data assets."

Slide 9: Methodological Limitations

Speaker : YOU

"As statisticians, we must acknowledge the limitations of our models. I optimized this pipeline for precision over recall to ensure these economic estimates act as a highly reliable floor, rather than an inflated ceiling.

There are three main caveats:

1. **Flow versus Stock:** Job adverts represent a *flow* of new hiring intent, not the total *stock* of the existing workforce. We must extrapolate proportional intensity.
2. **Internal Tools:** The AI struggles with proprietary internal company software. If a bank uses an internal database called 'Project Zeus', the AI won't recognize it unless it shares semantic traits with public tools.
3. **Conservative Thresholds:** My 3-term classification rule is strict. If a genuine data worker's advert was poorly written by HR and only mentions two technical skills, they are missed entirely."

Slide 10: Structural Economic Shifts

Speaker : YOU

"Despite these conservative limitations, when we apply this valuation framework, the macroeconomic impact is undeniable.

Here we are looking at the Total UK Gross Value Added compared against our Data Entry Investment Scenarios. Because the Alpha markup involves macroeconomic estimates, we provide a transparent confidence band—showing our conservative lower-bound scenarios alongside our upper-bound UK-specific markups.

What this chart clearly demonstrates is the sheer scale of investment happening just within this single occupation. It proves that 'own-account' data creation is a massive, structural component of the UK's overall GVA. Data capital formation is no longer just a niche tech industry output; it is a universally integrated foundation driving value across the economy.

To talk a bit more about how this pipeline will be utilized moving forward, I'll pass it back to

Stakeholder's Name

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Slide 11: What comes next? (NEW SLIDE)

Speaker : STAKEHOLDER

(Note for you: Let them speak. They will cover:)

- *Displaying this method at the ESCoE conference next week.*
- *Assessing this NLP method against the older subjective list of occupations.*
- *Which gives a better indication? Which is more robust and repeatable?*
- *MAC engagement.*

Slide 12: Conclusion & Q&A

Speaker : YOU

(You take the final wrap-up)

"Thank you. Just to close out before we take questions, this project proves that we can successfully turn massive, unstructured text datasets into a rigorous, objective valuation framework. It provides a real-time mechanism to track digital investment without relying on lagging surveys, and its architecture is fully adaptable for other labor market challenges.